Target recovery with matched endomorphism equations

The fixed 42-configuration panel has independently verified native/PARI results. Twenty-nine bases fill; thirteen remain underfilled. Of 252 planned matrix cases, 174 are analyzed and 78 are excluded by the predeclared underfill rule. Adding known endomorphism equations resolves 14 previously ambiguous cases. Ordinary bases benefit as well as twist-derived bases. This is one public toy DLP with correlated matrix cases, not 114 independently recovered targets or a cost result.

Base rule and transform	Analyzed off/on pairs	Target fixed, off	Target fixed, on	Newly fixed
Curve progression, half seeds plus alpha images	15	8	13	5
Curve progression, plain	18	12	12	0
Ascending x, half seeds plus alpha images	9	6	8	2
Ascending x, plain	9	6	6	0
Maximal-seed twist progression, half seeds plus alpha images	18	10	16	6
Maximal-seed twist progression, plain	18	8	9	1

These are recorded exact counts from `execution-v14/SUMMARY.json`. The 14-case change is the sum of its newly-fixed counts, $5 + 2 + 6 + 1$. Unequal denominators, overlapping arity modes and reused target forbid treating this table as a population success-rate comparison. The first-seed twist rules are absent from the filled table because all twelve underfill; their full records remain.

Exact task and controls

The unchanged source is $E : y^2 = x^3 + x + 15$ over $\mathbb{F}_{827}$, order823, $G = (2, 5)$, ICV1 `icv1-fp10-t5-192cb216`; public $Q = [37]G$. Its rational twist is $T : Y^2 = X^3 + 4X + 120$, order833, group $C_7 \times C_{119}$. The given first seed $(0, 51)$ has order7 and maximal seed $(2, 386)$ has order119. The certified degree17 map $\alpha = (\pi + [1])/[7]$ acts as353 on $E(\mathbb{F}_{827})$. Map discovery was an input to this diagnostic and is not charged here.

For each B8/12/16, seven fixed rules use ascending x, curve progressions at shifts1/3, and first/maximal twist progressions at shifts1/3. Orbit coordinates are lifted on the original curve after the fixed shift; twist X is divided by2 first. G anchors each base. The plain transform requests B seed points; the other requests B/2 and unions them with canonical alpha images. There is no post-outcome refill, replacement, truncation or base selection using Q or its log.

All ordinary and twist bases receive the same map checks and known scalar information. Internal canonical edges give $u_j - cu_i = 0 \pmod{823}$ with $c = 353$ or -353, depending on orientation. Compare these equations off/on. An internal edge is known information, not a newly sampled query relation.

Every filled base collects all signed two-/three-term witnesses for $Q + [j]G$, $j = 0, \ldots, 15$, including repetitions and multiplicities. Each base has arity2, arity3 and their union, crossed with equations off/on. The single native execution retains 928 attempted queries, 402 zero-witness queries and 5389 raw witnesses. The remaining planned queries were not attempted on underfills. All395 base-point

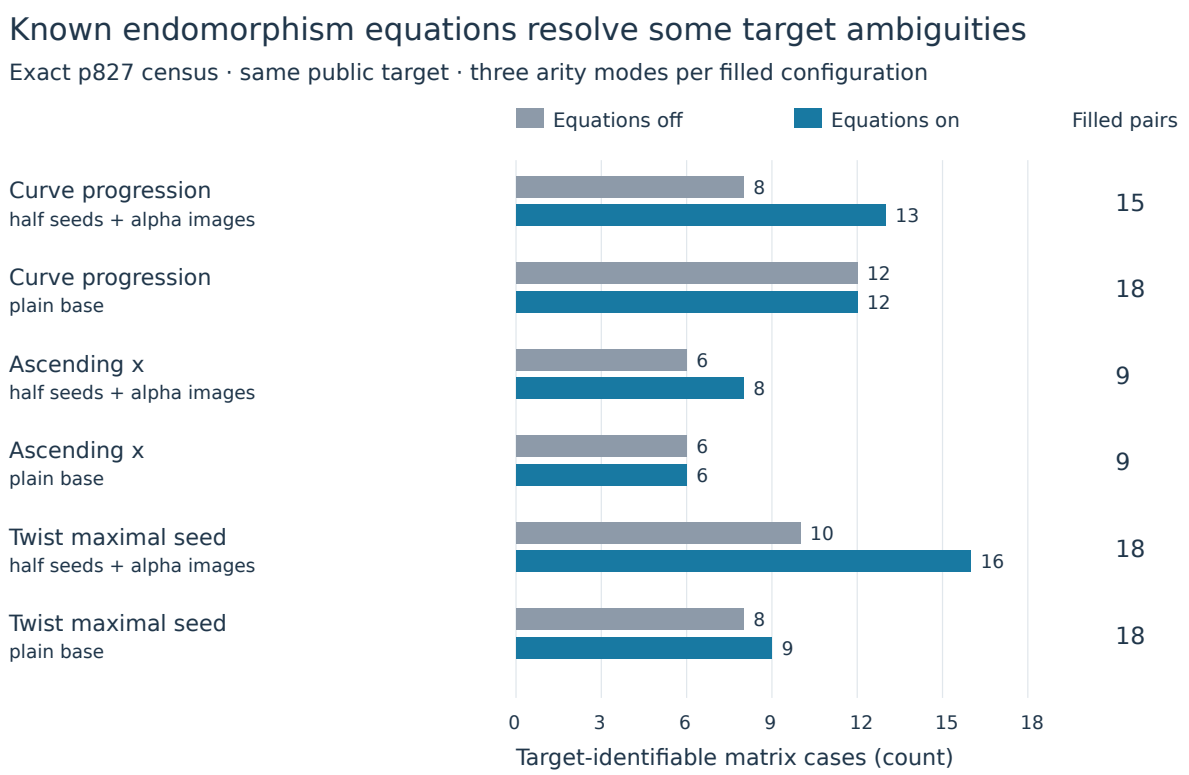


Figure 1: Recorded target-identifiability counts with known alpha equations off/on. Native run rank-v1 and independent GP run pari-v1, protocol v14. Counts are a complete finite census; no sampling uncertainty or clocks are admitted.

map/scalar controls pass, including rejection of scalar354. Two archived bases and all64 corresponding query witness sets match v8 exactly.

All twelve first-seed twist configurations underfill: their final sizes are 3/6 for shift1 plain/augmented and1/2 for shift3. The remaining underfill is B16 curve-shift1 augmented, final size15 after a collision. All13 records and their78 excluded matrix cells are explicit. No extra queries were sampled.

What changed, and what stayed ambiguous

The archived B8 maximal-twist shift1 plain base still gives66 raw witnesses, 15 distinct equations, rank8 in9 unknowns, and an ambiguous target. It has no internal alpha edge, so enabling the equations changes nothing.

For B8 maximal-twist shift3 with half seeds plus alpha images, arity3 gives 19 raw witnesses and8 distinct query equations. Known equations raise rank6 to9 in9 unknowns and determine the public target scalar37. This verifies one modified twist-derived base that resolves the earlier type of ambiguity; it does not repair the original base or prove a twist-specific mechanism.

The ordinary B8 curve-shift3 augmented base gives an informative counterpoint: arity2 has3 distinct query equations; adding known equations raises rank3 to7 in9 unknowns, and the target is fixed although other unknown logs remain free. The maximal-twist B16 shift1 plain arity2 case also fixes the target at rank12 in17 unknowns. Thus full column rank is sufficient but unnecessary for recovery.

The analysis keeps the rank of anchor/alpha equations separate from the new rank contributed by query equations. It does not credit known scalar edges as independent sampled relations. Across all174 analyzed cells, 114 fix the target and60 leave it ambiguous; every fixed scalar passes the original public point check. Every ambiguous case includes two distinct target coordinates satisfying every equation. The eliminator receives no factor-base log table or scalar37.

Independent and formal verification

For $Mx = b$, a particular solution x_0 and a complete kernel basis describe the affine fiber $x_0 + \ker M$. The target coordinate is fixed exactly when its projection vanishes on the whole kernel. Native exact RREF records this fiber, not just rank. PARI independently rebuilds every base and alpha map using `ellisogeny`, every edge, witness, deduplicated equation, rank, kernel dimension and span, particular solution, target projection and recovery/ambiguity check.

Both scientific processes complete successfully. Strict admission checks process status, all script/output hashes, sole marker `ALL_TWIST_RANK_REPAIR_OK 42`, and empty stderr or only the predeclared GP stack-size warning. The combined export has two planned, two attempted and two completed processes; zero interrupted, resumed or excluded processes. Base underfills are within-run exclusions, not missing process rows.

In namespace `TwistTargetIdentifiability`, the Lean declarations `target_fiber_iff_kernel` and `kernel_witness_changes_target` prove the general additive-homomorphism fiber criterion and an explicit ambiguity witness. Lean4.34.1, pinned mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612`, frozen build and fresh native transitive axiom audit pass for both theorems, standard foundations only. The concrete finite matrix, field, complete kernel basis and elliptic-curve extraction bridge remain outside these formal proofs. This is not a formally checked proof of every concrete matrix. No characteristic827 symbolic identity is being promoted through the fixed-modulus identity-certificate format.

The native diagnostic's19 tests and full standalone runner package pass. Before freeze, synthetic F101 controls exposed two GP matrix-shape defects; both failed attempts are preserved, followed by a successful corrected control. The first Lean development import failure is retained with the successful development/frozen replays. The main panel ran once with no retry/amendment. The earlier named full-root validation gap remains open; this package pass does not imply a complete root suite or CI pass.

Pins, reproduction and scope

Source `d377251611bc9fd3da22096c3b128862816ef268`; committed freeze before data; protocol hash `e031cee91eef30ca50d2ddef24cd36c7ca6ee739a3f173bd9885005efb7c61bf`. Inputs, binaries, commands and caps are in execution-v14 FROZEN/PROTOCOL files. Native and GP use one thread, CPU0, 4GiB and300s each on the managed Xeon8573C KVM host, L0, under the recorded EKS image-access fallback. No clocks, paid allocation or external-tool performance comparison is claimed. The controls use independent implementations on the same host/author, not independent review.

Raw archive:61888bytes/ten files, SHA256 `2fd7a36e7a9755ea9687e93d7cc86bec661943dcf6fb6385 3968432b52d41e13`. It is in the user-confirmed `s3://setec-astronomy-artifacts/store`; publication verification records the fresh download, all member hashes and deterministic repack separately. See execution-v14 for commands, the complete42/252-cell records, unified export and exact validation receipts.

Literature reviewed before implementation: repository factor-base/GLV synthesis, v8/v9 rank evidence and v10 map construction; primary MIT18.783 Fall2025 Lecture13, Theorems13.6/13.8; official PARI matrix documentation and installed help for `matker`, `matrank`, `matinverseimage`. Affine-kernel linear algebra and endomorphism scalar equations are standard methods.

This completes the bounded relation-repair diagnostic in its filed card, including the matched augmentation extension. It leaves the separately frozen complete cold-cost panel, strongest ordinary/group/external controls, larger held-out curves, original E00–E29 campaigns and concrete formal bridges open. The result provides no novelty, IC speedup, asymptotic or NIST conclusion. Canonical progress timeline, IC scoreboard, curve diagrams and performance charts were checked: no model/map or admitted cost value changes, so those graphs are unchanged. The new quantitative graph records this diagnostic only.